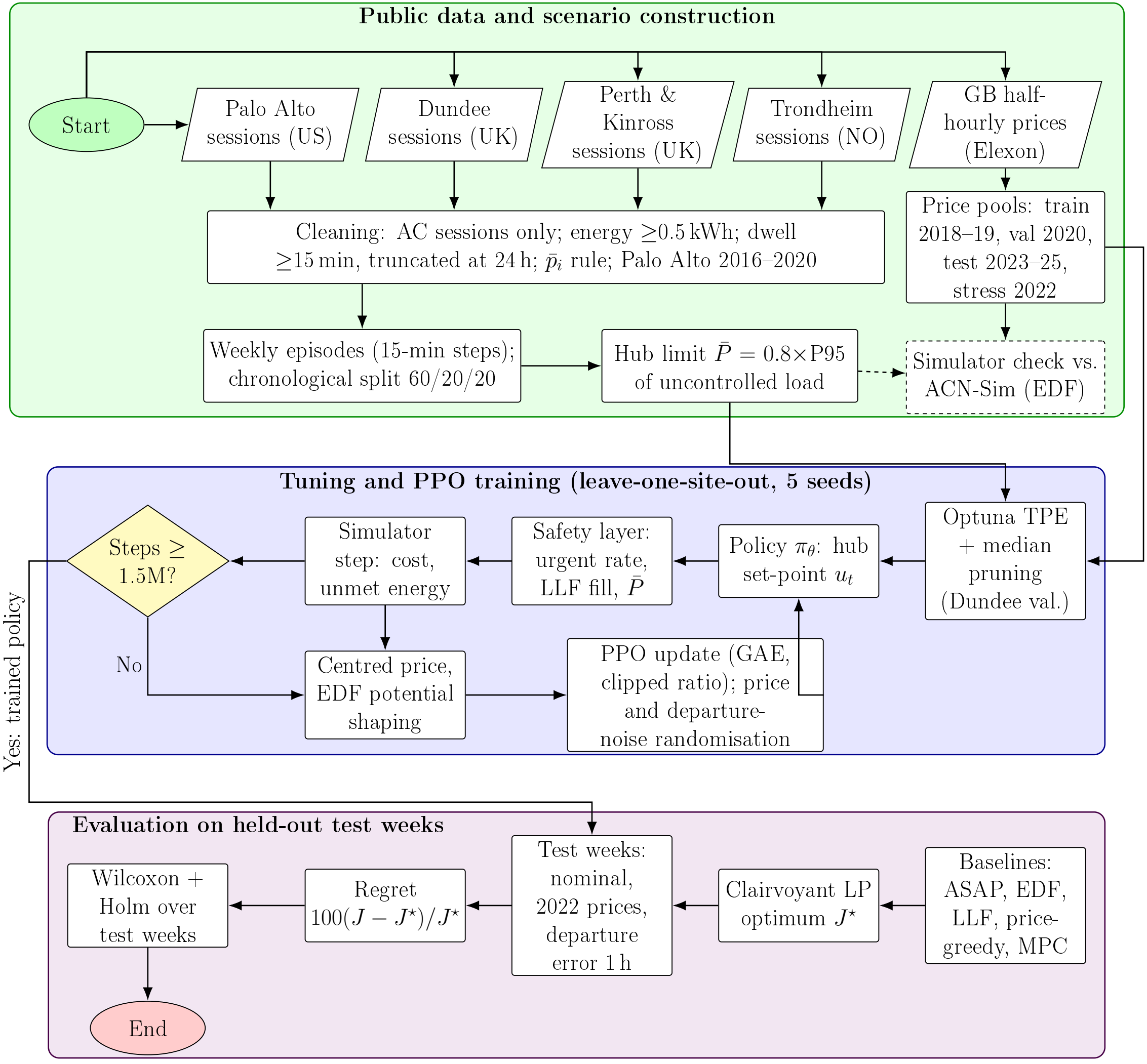

Public data and scenario construction
Start
Palo Alto sessions (US)
Dundee sessions (UK)
Perth & Kinross sessions (UK)
Trondheim sessions (NO)
GB half-hourly prices (Elexon)
Cleaning: AC sessions only; energy ≥0.5 kWh; dwell ≥15 min, truncated at 24 h; $\bar{p}_i$ rule; Palo Alto 2016–2020
Price pools: train 2018–19, val 2020, test 2023–25, stress 2022
Weekly episodes (15-min steps); chronological split 60/20/20
Hub limit $\bar{P} = 0.8\times$P95 of uncontrolled load
Simulator check vs. ACN-Sim (EDF)
Tuning and PPO training (leave-one-site-out, 5 seeds)
Optuna TPE + median pruning (Dundee val.)
Policy π_θ: hub set-point u_t
Safety layer: urgent rate, LLF fill, $\bar{P}$
Simulator step: cost, unmet energy
Steps ≥ 1.5M?
No
Yes: trained policy
Centred price, EDF potential shaping
PPO update (GAE, clipped ratio); price and departure-noise randomisation
Evaluation on held-out test weeks
Test weeks: nominal, 2022 prices, departure error 1 h
Clairvoyant LP optimum $J^\star$
Baselines: ASAP, EDF, LLF, price-greedy, MPC
Regret $100(J - J^\star)/J^\star$
Wilcoxon + Holm over test weeks
End